

To-Do List

1. Implement missing strategies

There are two strategies remaining to be implemented (see Scenarios 3 and 4 of the Scenarios report).

- **Class: CloneFusionStrategy.** The principle is to search for various "clones" in the graph by calling CloneIdentifySpecialist, then to compress them using CloneFusionSpecialist. After performing the isomorphism search to retrieve the global solution, use DecompressionSpecialist.
 - Implementation steps:
 - Apply CloneIdentifySpecialist to identify symmetric arguments (clones) in the graph.
 - Use CloneFusionSpecialist to merge these clones into a meta-argument.
 - Search for an isomorphic case to the compressed graph in the case base via IsomorphismSpecialist.
 - Apply BijectionSpecialist to map the nodes, then use DecompressionSpecialist to reintegrate the fused arguments into the final solution.
- **Class: DirectionalCBRStrategy.** The principle is to answer a local question regarding a specific argument by isolating the relevant subgraph.
 - Implementation steps:
 - Apply DirectionalPruningSpecialist to prune non-cyclic successors of the target argument.
 - Search for an isomorphic case to the pruned graph in the case base via IsomorphismSpecialist.
 - Apply BijectionSpecialist to transfer the acceptance status of the extracted solution to the current problem.

2. Hierarchy of questions

Currently, we search for isomorphic cases where the entire problem (graph and question) is isomorphic. However, the isomorphism search should ideally be performed only on the graph to allow the reuse of cases that have different questions but may still be useful for our target case. The idea is to create a hierarchy of questions to facilitate processing.

3. Filters for isomorphism

Search Currently, the only filter in place is sorting by the graph's hash value. We could imagine other filters to limit costly comparisons.

4. Add unimplemented parts related to the SAT solver

These include the studies of credulously and sceptically accepted arguments, and the "from-scratch" resolution strategy if no CBR approach succeeds.

5. Strategy workflow

Currently, a strategy is a simple sequence of specialists. This should be modified so that a strategy makes recursive calls on a list of specialists, while ensuring that a specialist is not called on a problem it has already processed.